

Daily Problem: 05–Feb–2026

Problem Statement

Fix a finite alphabet Σ equipped with a total order $\leq_\Sigma$. This induces the usual lexicographic order $\leq$ on Σ^* . For a language $L \subseteq \Sigma^*$, define its *lexicographic downward closure* by

$$\downarrow L = \{ w \in \Sigma^* \mid \exists u \in L \text{ with } w \leq u \}.$$

Show that if L is regular, then $\downarrow L$ is also regular.

Proof

Let L be regular. Then there exists a DFA

$$M = (Q, \Sigma, \delta, q_0, F)$$

such that $L = L(M)$.

Lexicographic order: a useful characterization

For words $w, u \in \Sigma^*$, we have $w \leq u$ in lexicographic order if and only if one of the following holds:

1. w is a prefix of u , i.e. $u = wv$ for some $v \in \Sigma^*$, or
2. there exist words $x \in \Sigma^*$ and letters $a, b \in \Sigma$ and suffixes $y, z \in \Sigma^*$ such that

$$w = xay, \quad u = xbz, \quad \text{and} \quad a <_\Sigma b,$$

and x is the *first* position where w and u differ.

This is exactly the usual “first differing symbol decides” rule.

Goal restated

A word w is in $\downarrow L$ iff there exists some accepted word $u \in L$ such that u is either:

- an extension of w (so w is a prefix of an accepted word), or
- agrees with w up to some position, then uses a *strictly larger* letter than w at the first differing position, and then continues arbitrarily to reach acceptance.

We will build an NFA that checks exactly this condition, and then conclude regularity.

Construction of an NFA for $\downarrow L$

We construct an NFA N that reads an input word w and nondeterministically guesses how an accepted word u witnessing $w \leq u$ looks.

The NFA has two modes:

- **Mode 0 (equal-so-far):** we are still matching w and the witness u symbol-by-symbol (so far they are equal).
- **Mode 1 (already-smaller):** we have already ensured $w < u$ at some earlier position (by choosing a larger letter in u), so from now on we only need to check that u can be completed to some accepted word in L , regardless of what symbols w continues to have.

Formally, define the NFA

$$N = (Q', \Sigma, \Delta, s, F'),$$

where:

- $Q' = (Q \times \{0, 1\})$. (We track a state of the DFA M and the mode.)
- The start state is $s = (q_0, 0)$.
- The accepting states are

$$F' = \{(q, 0) : q \in \text{Acc}\} \cup \{(q, 1) : q \in \text{AccReach}\},$$

where:

- $\text{Acc} = F$,
- $\text{AccReach} = \{q \in Q : \exists v \in \Sigma^* \text{ with } \delta^*(q, v) \in F\}$,

i.e. AccReach is the set of states from which some accepting state is reachable in M . (This is a standard DFA graph reachability notion.)

Transitions

Let $c \in \Sigma$ be the next input symbol of w . The NFA guesses how the witness word u behaves.

(1) Staying equal-so-far (Mode 0). If we keep u equal to w at this position, then the DFA M would read the same symbol c . So we include:

$$(q, 0) \xrightarrow{c} (\delta(q, c), 0).$$

(2) Making w strictly smaller at the first differing position (switch to Mode 1). At some point, we may decide this is the first position where u exceeds w . If the input symbol is c , we may choose any symbol $d \in \Sigma$ with $d >_{\Sigma} c$ for the witness word u . This is a nondeterministic choice made by the automaton. If we choose such a d , then M would transition by d , and we switch to Mode 1:

$$(q, 0) \xrightarrow{c} (\delta(q, d), 1) \quad \text{for every } d \in \Sigma \text{ with } d >_{\Sigma} c.$$

(3) After $w < u$ has been established (Mode 1). Once we are in Mode 1, the witness u may use any symbols at all while still aiming to reach acceptance. Since the NFA must consume the input symbol c (which belongs to w), it is free to choose *any* symbol $d \in \Sigma$ as the corresponding symbol of u . Thus we include:

$$(q, 1) \xrightarrow{c} (\delta(q, d), 1) \quad \text{for all } d \in \Sigma.$$

(4) Handling the “prefix” case w is a prefix of u . If w is a prefix of u , then after reading all of w we must be able to extend the current DFA state to an accepting state. This is exactly why we made all states $(q, 0)$ with $q \in \text{AccReach}$ accepting in Mode 0 *only at the end of input*. Concretely, at the moment the input finishes, acceptance depends on whether the current DFA state q can reach acceptance.

This is captured by including $(q, 0)$ as accepting whenever $q \in \text{AccReach}$ (and similarly in Mode 1).

Correctness

We prove that $L(N) = \downarrow L(M)$.

($\subseteq$) If N accepts w , then $w \in \downarrow L$.

Suppose N accepts $w = c_1 c_2 \cdots c_m$. Consider an accepting run of N . At each step, the run updates a DFA-state component from Q . By the definition of transitions, this run corresponds to a sequence of guessed symbols $d_1, \dots, d_m \in \Sigma$ that M would read, where:

- in Mode 0, $d_i = c_i$ unless the run takes a switch transition with $d_i > c_i$,
- once Mode 1 is entered, each d_i can be arbitrary.

Let $u_0 = d_1 \cdots d_m$. If the run never switches to Mode 1, then $u_0 = w$. If it switches at some first position i , then u_0 agrees with w on positions $< i$ and has $d_i > c_i$ at the first difference, hence $w < u_0$, i.e. $w \leq u_0$.

Now, because the run is accepting, the final DFA-state component q lies in AccReach . Therefore there exists some suffix $v \in \Sigma^*$ such that $\delta^*(q, v) \in F$. Define $u = u_0 v$. Then $u \in L(M) = L$ and $w \leq u$. Hence $w \in \downarrow L$.

($\supseteq$) If $w \in \downarrow L$, then N accepts w .

Assume $w \in \downarrow L$. Then there exists $u \in L$ with $w \leq u$. Let $w = c_1 \cdots c_m$. Consider two cases.

Case 1: w is a prefix of u . Then $u = wv$ for some v . Running M on w lands in some state q from which acceptance is reachable (since continuing with v reaches F). Thus $q \in \text{AccReach}$, so the NFA can stay in Mode 0 throughout and accept at the end.

Case 2: $w < u$ at a first differing position. Then there exists an index $i \leq m$ such that $c_1 \cdots c_{i-1} = u_1 \cdots u_{i-1}$ and $c_i <_\Sigma u_i$. The NFA can stay in Mode 0 reading $c_1, \dots, c_{i-1}$ and matching the same symbols in M . At position i , it takes the switching transition using $d = u_i$ (which is $> c_i$), entering Mode 1. After that, it continues reading the rest of w while guessing the corresponding symbols from u 's continuation (or any other symbols), thus simulating a run of M along some word with prefix matching the witness. Since $u \in L$, the resulting DFA-state at the end is in AccReach , so the NFA accepts.

In either case, N accepts w .

Combining both directions, $L(N) = \downarrow L$. Hence $\downarrow L$ is regular.

Conclusion

Starting from a DFA M for L , we constructed an NFA that guesses a witness $u \in L$ satisfying $w \leq u$ while reading w . By closure of regular languages under NFA/DFA equivalence, it follows that $\downarrow L$ is regular whenever L is regular.